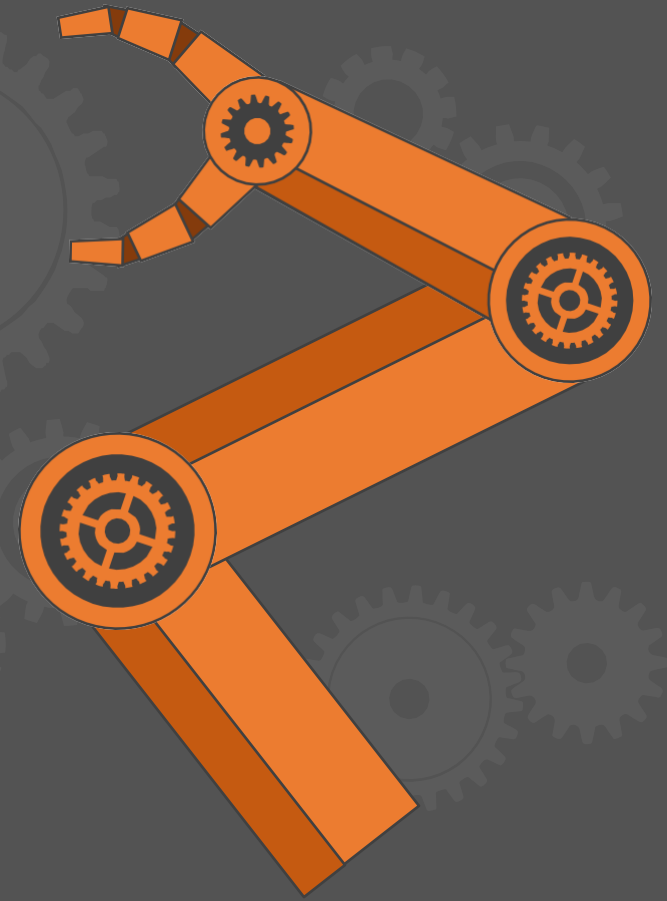
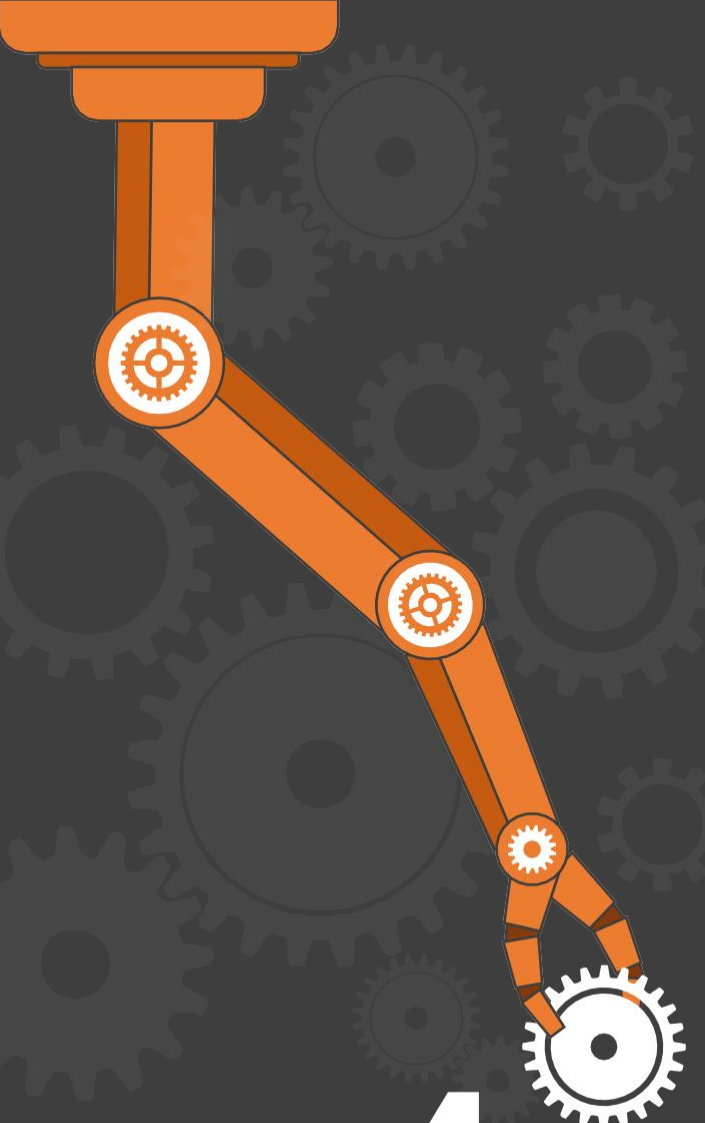


PARKING LOCK RAD LINE ASSEMBLY LINE LAYOUT





4. INDUSTRY

Proposed Methodology

➤ Feeding of the Parking Lock product in OP10

- Operator manually feeds the component to the Robot for the OP10 machine.
- The Cobot-1 feeds the component to the OP10 machine. Two OP10 machines are used in this line. The cycle time is 67sec per machine
- The output of the OP10 is feed to the conveyor for the OP20 process

➤ Transfer to OP-20 Machine

- Conveyor-1 transfers the object to OP20 Machine
- Cobot 2 pick the component from Conveyor 1 and feeds to the OP20 machine. Two OP20 machines are used in this line. The cycle time is 68 sec per machine.
- After OP-20 operation is completed, Cobot 2 places the object on Conveyor-2.

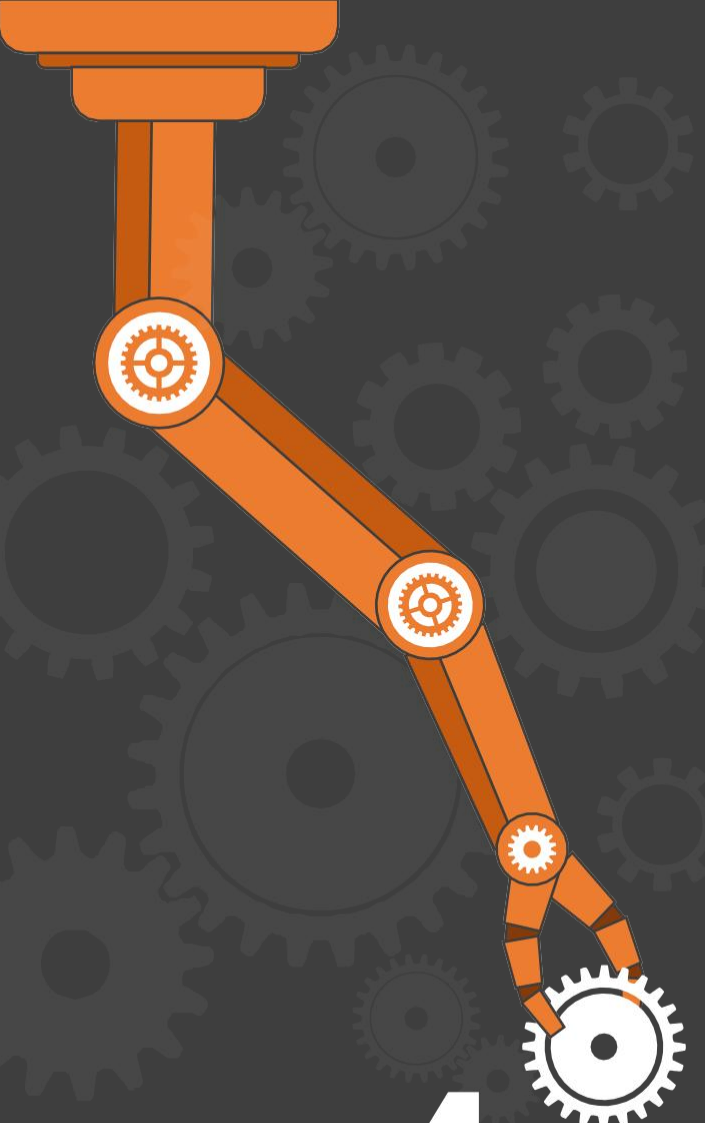
➤ Transfer to OP-30 Machine

- Conveyor-2 transfers the object to its other end.
- Cobot-3+AMR-1 picks the object and loads it into the OP-30 machines.
- After OP-30 operation is completed, Cobot-3+AMR places the object into the bin which is kept on pallet. The cycle time is 390 sec per machine.

Proposed Methodology

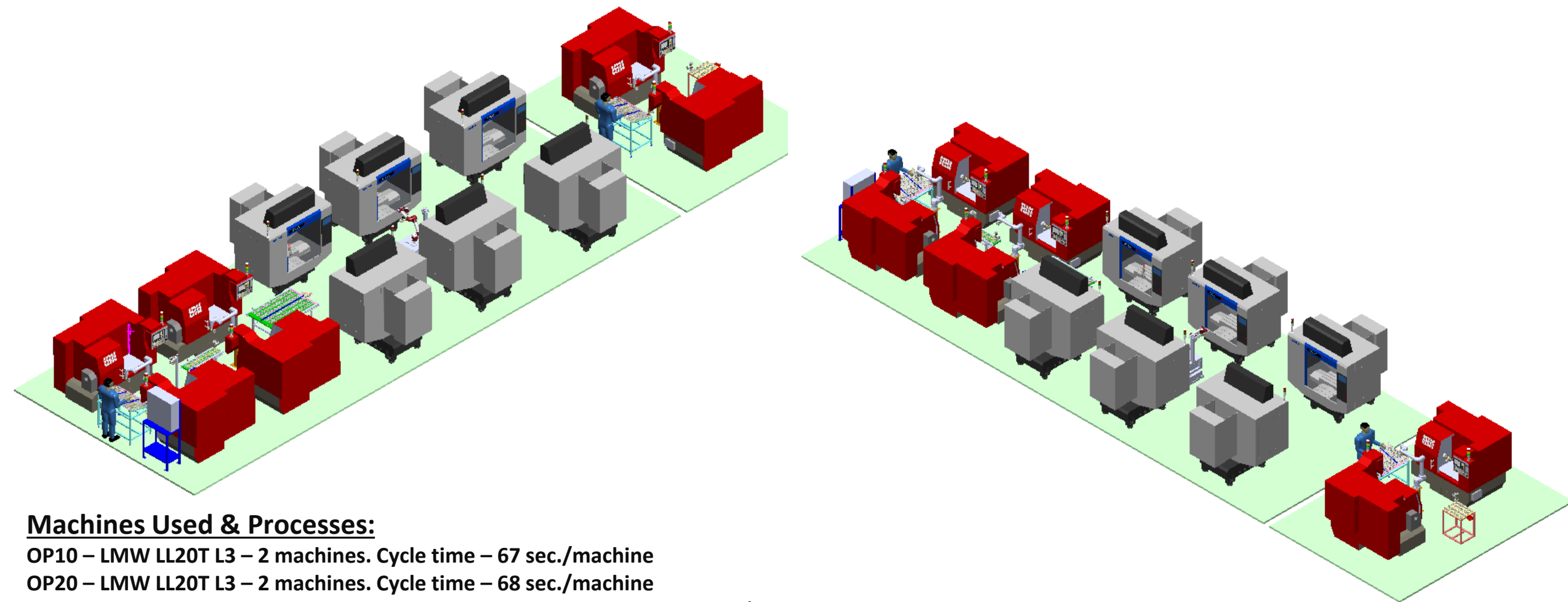
➤ Transfer to OP-40 Machine

- After OP-30 operation, the filled pallet is picked feed to OP40 and is picked up by the Cobot 4 and taken to the OP-40 Machine.
- In OP-40 Machine, the Cobot-4 picks from bin and feeds to OP-40 machine.
- Two sets of OP-40 machine are used in this line with the cycle time of 115 sec. each.
- The output is kept into the output bin which is kept on the conveyor/pallet.



4. INDUSTRY

Parking Lock RAD Line Assy Line Automation - Iso View



Machines Used & Processes:

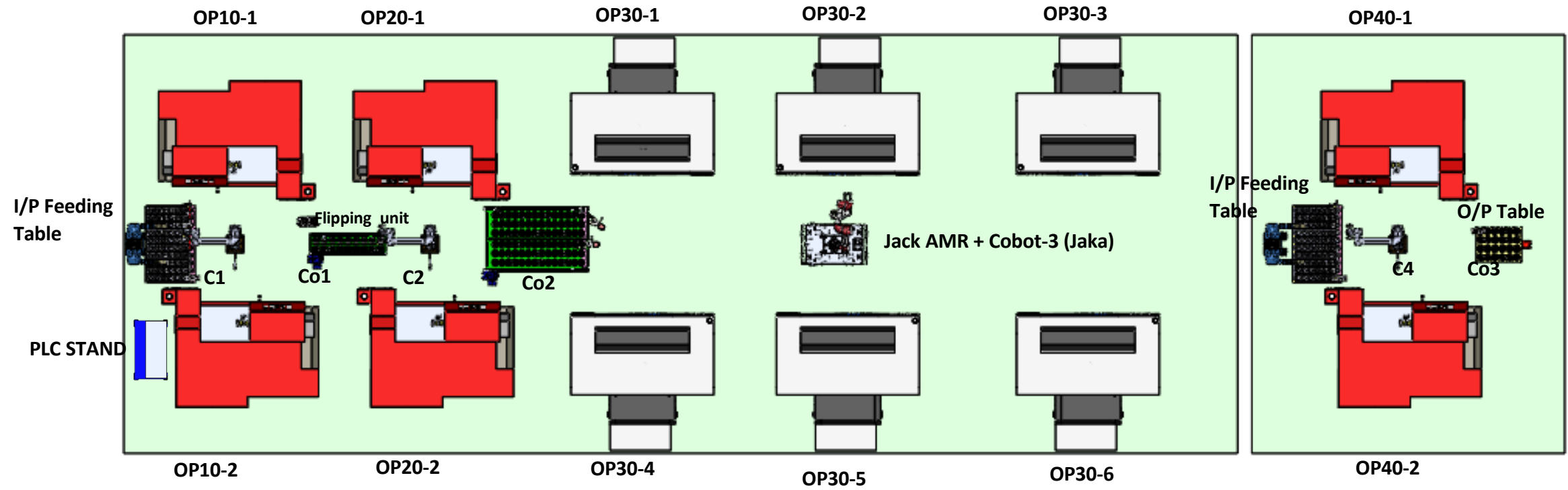
OP10 – LMW LL20T L3 – 2 machines. Cycle time – 67 sec./machine

OP20 – LMW LL20T L3 – 2 machines. Cycle time – 68 sec./machine

OP30 – LMW JV55 – 1 machine; JV Kraft – 2 machines. Cycle Time – 390 sec./machine.

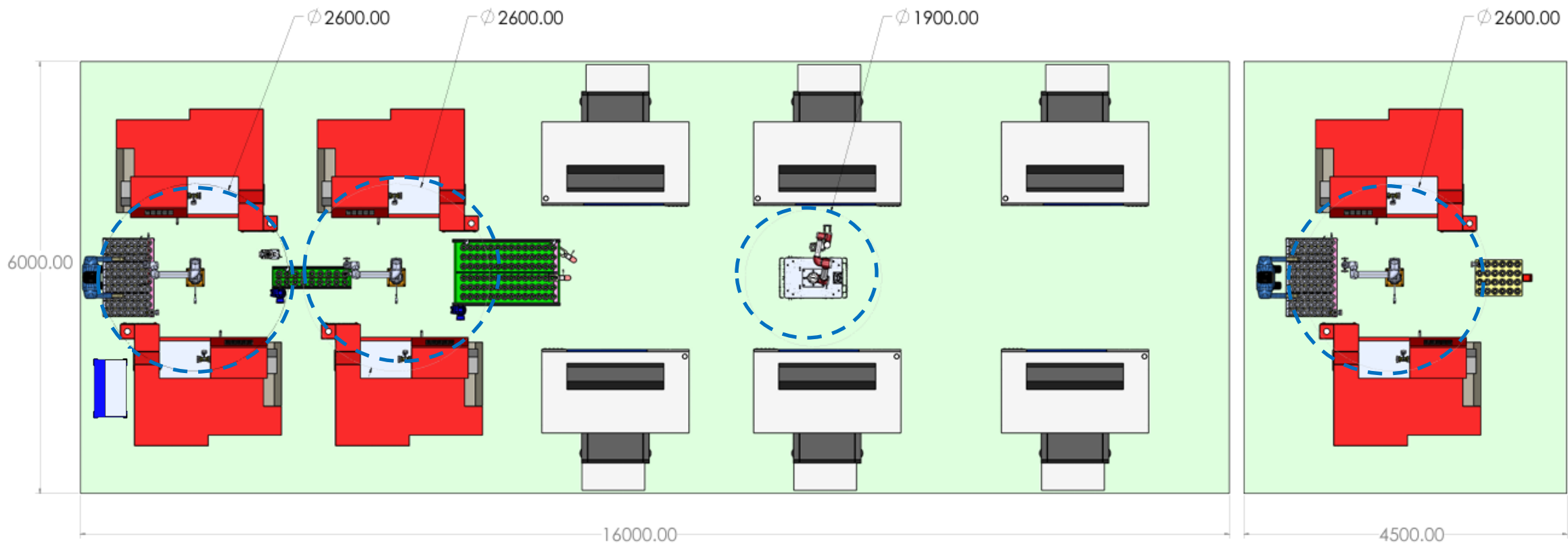
OP40 – 2 machines. Cycle Time – 115 sec.

Parking Lock RAD Line Assy Line Automation – Top View

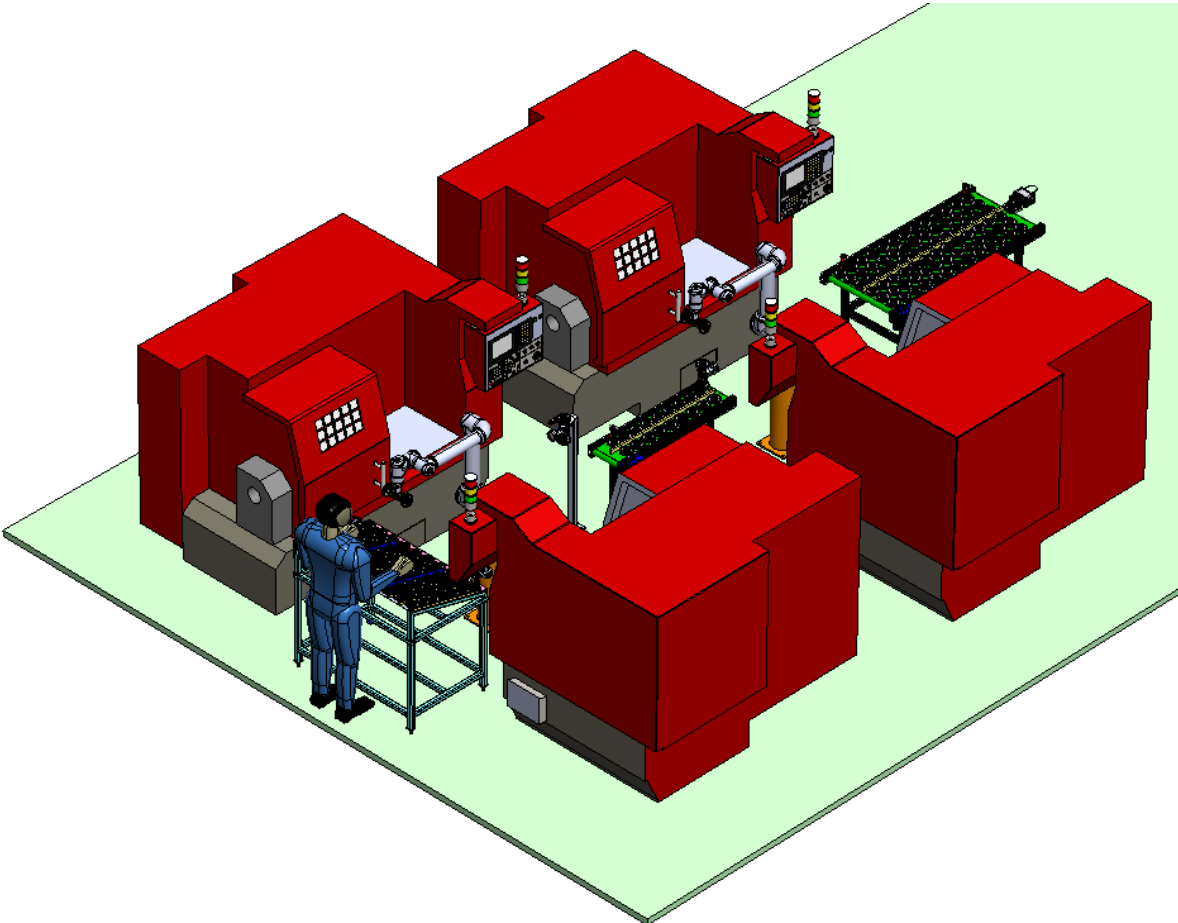
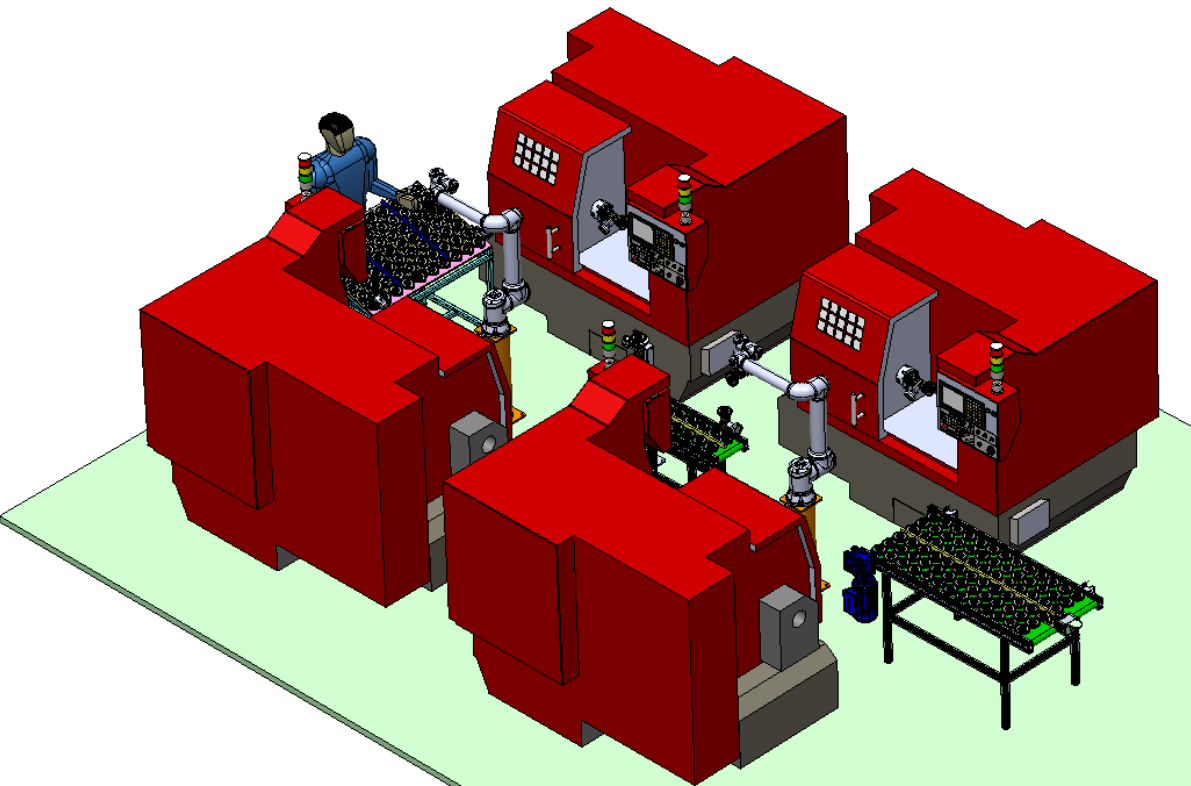


- C1 – Cobot 1
- C2 – Cobot 2
- C4 – Cobot 4
- Co1 – Conveyor 1
- Co2 – Conveyor 2
- Co3 – Conveyor 3

Parking Lock RAD Line Assy Line Automation – Robot Reach

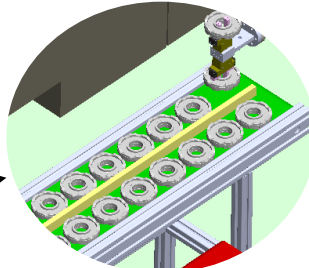
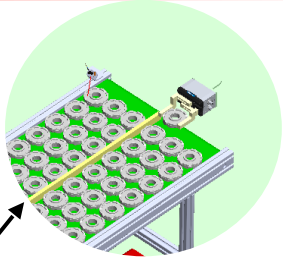


RAD Line–OP10&OP20- Iso View

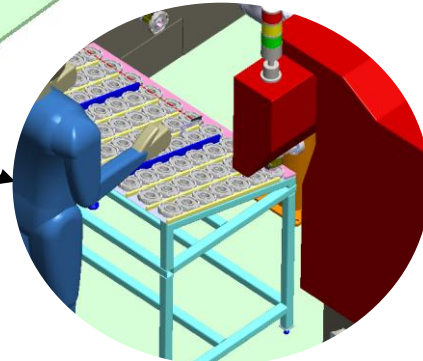


RAD Line–OP10&OP20- Gripper Dimension

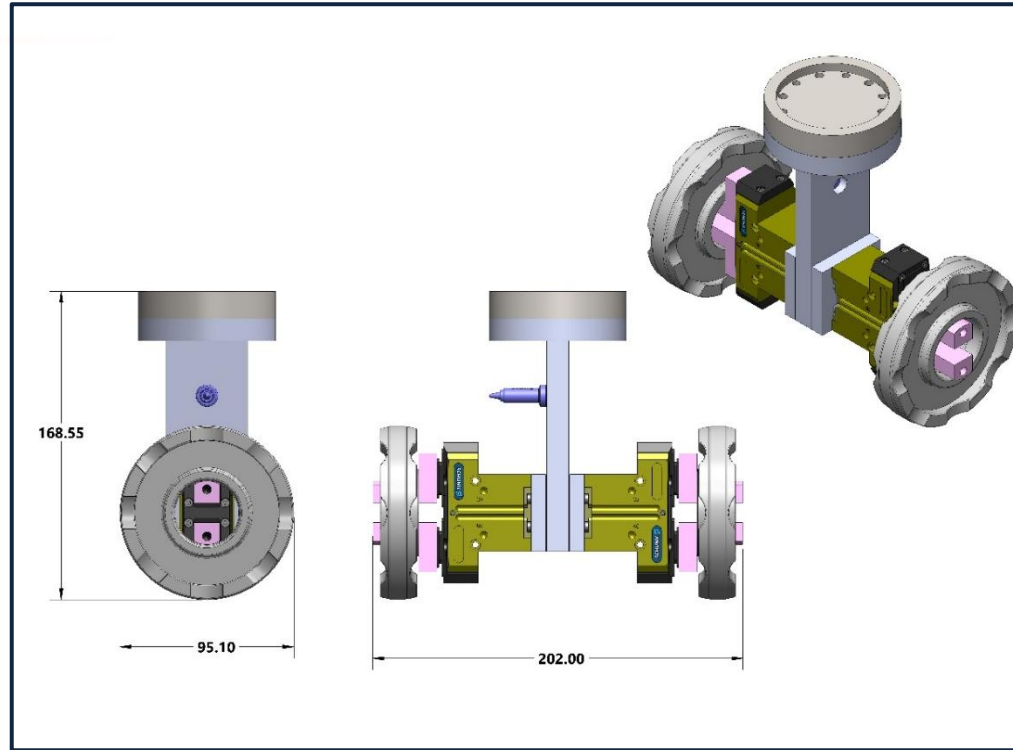
OP 20-30 Conveyor
Quantity on Conveyor : 78Nos



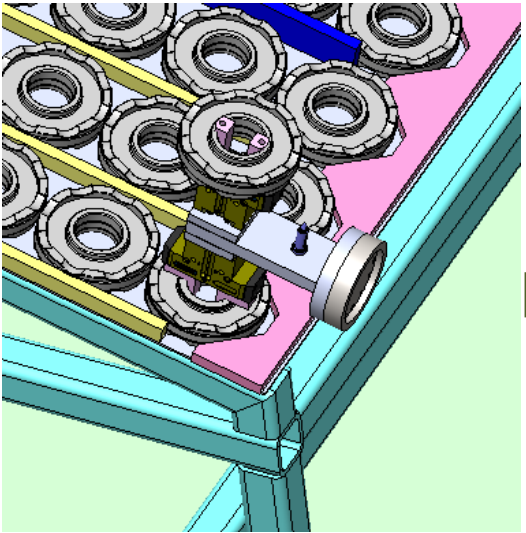
OP 10-20 Conveyor
Quantity on Conveyor : 18Nos



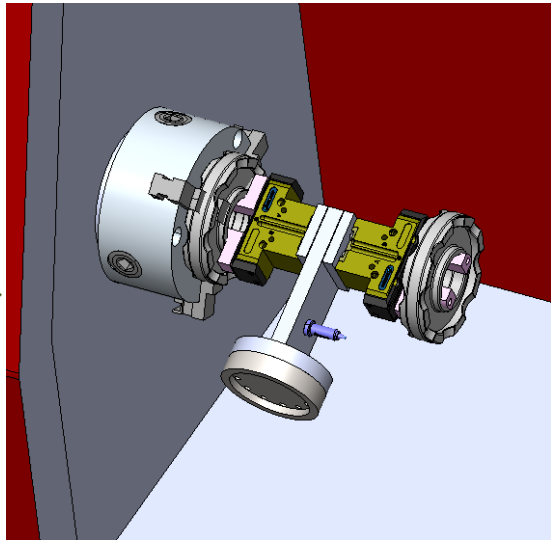
Input Loading –
Quantity on table : 70Nos



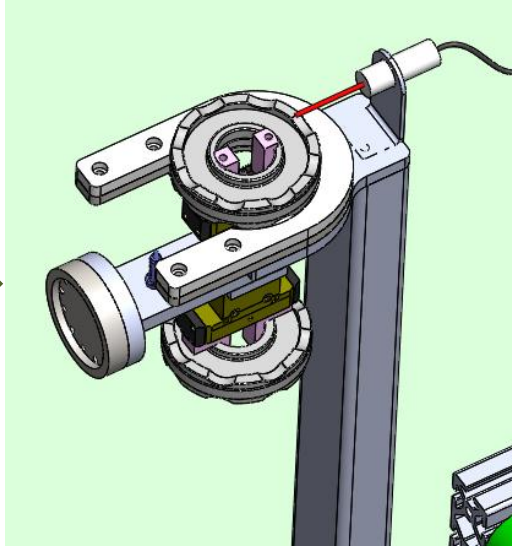
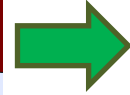
**RAD Line–OP10&OP20 -
Gripper Position**



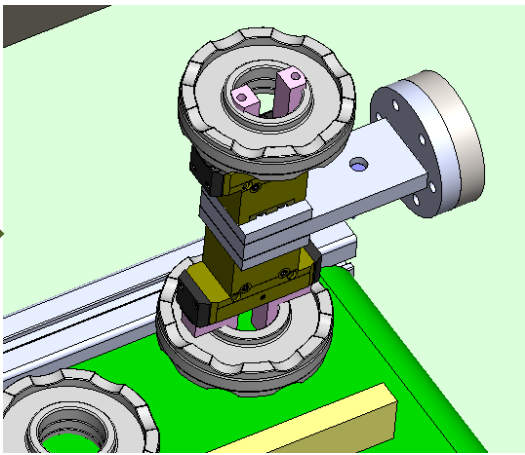
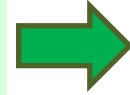
OP10 - PICKUP



OP10 - LOADING

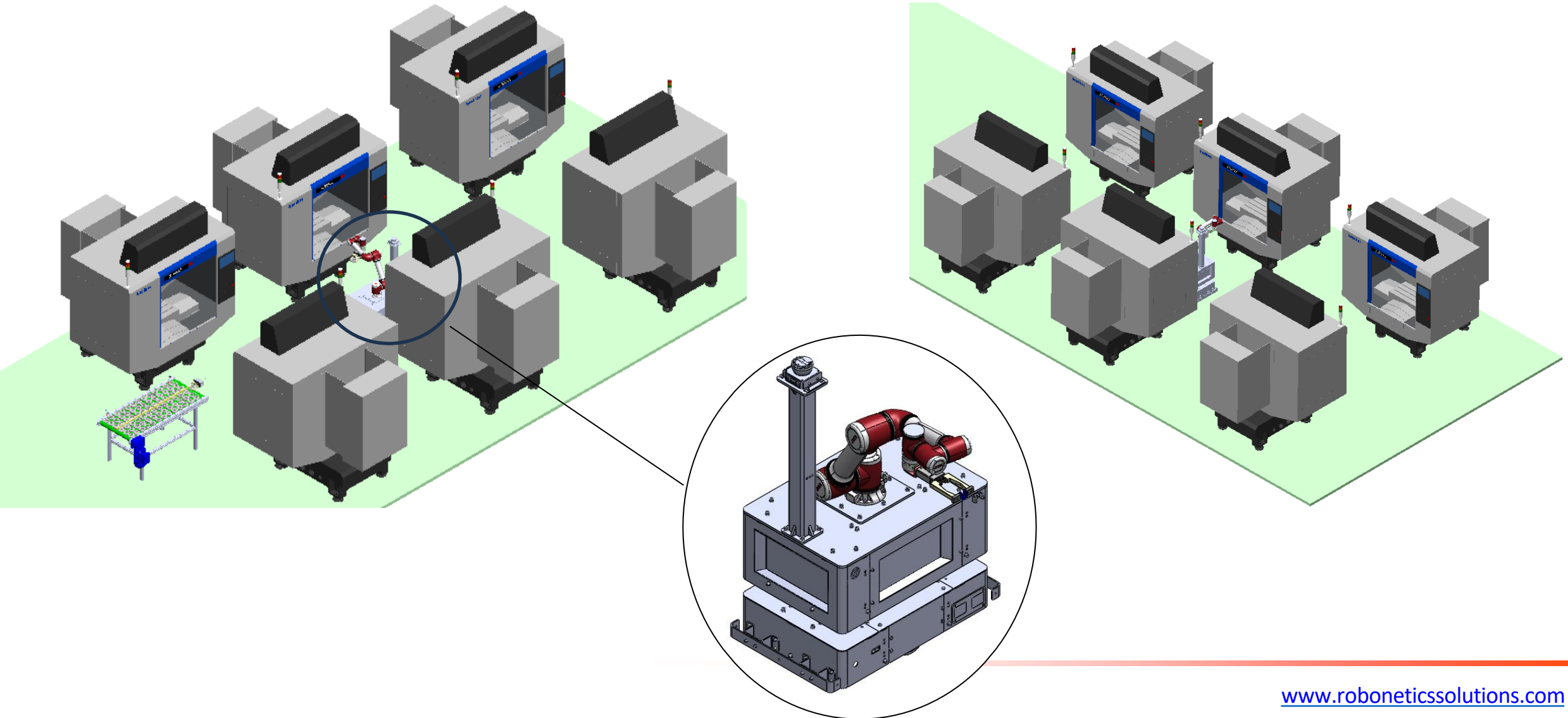


**Completed OP10 –
FLIPPING**

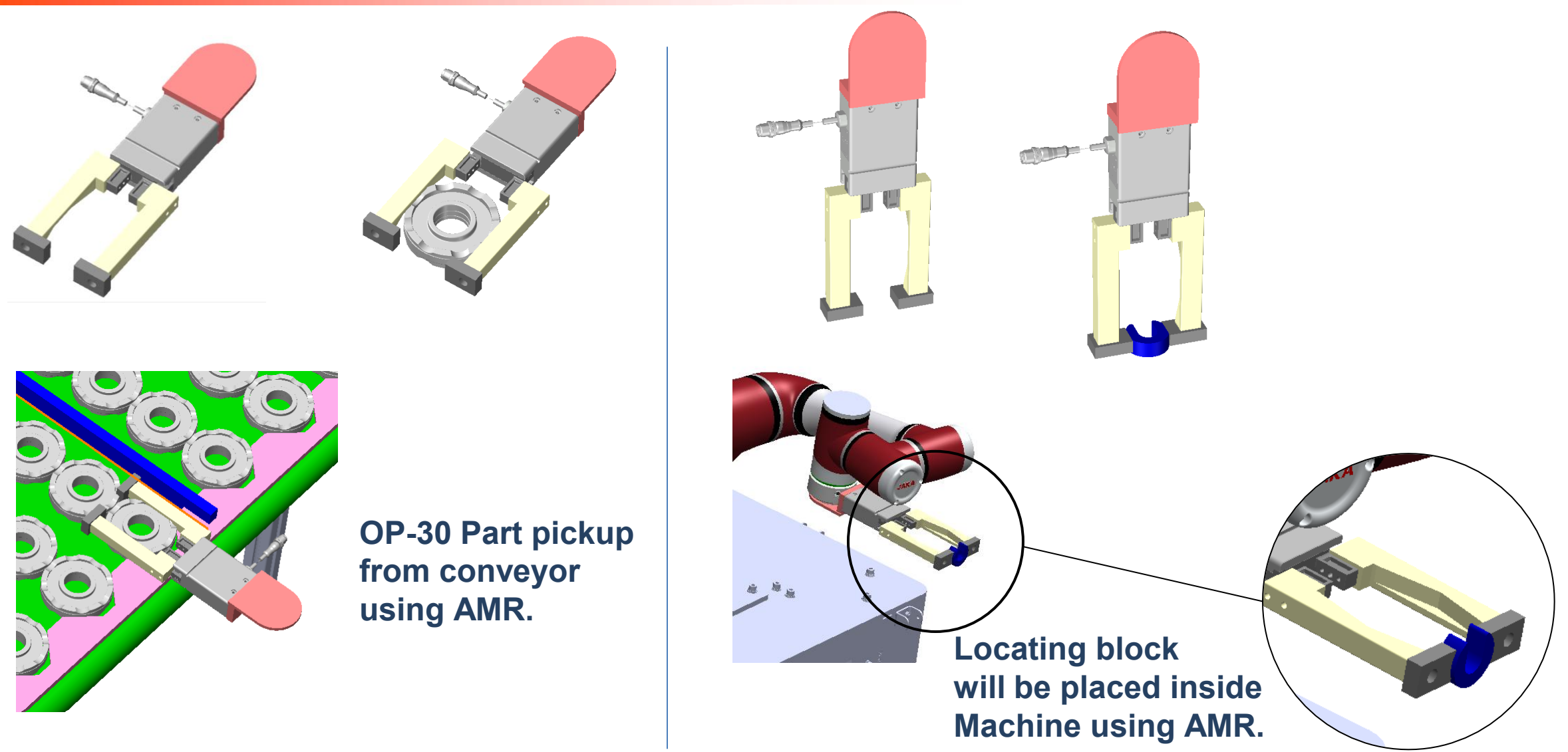


**Completed OP10 –
LOCATING IN
CONVEYOR**

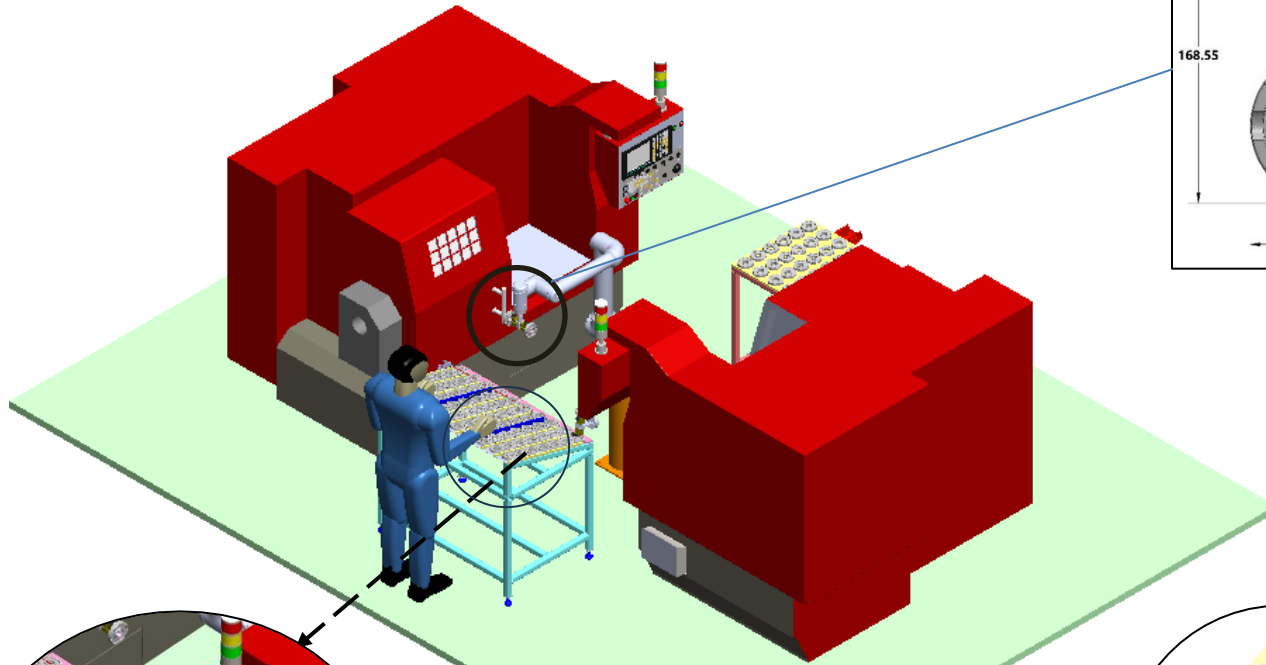
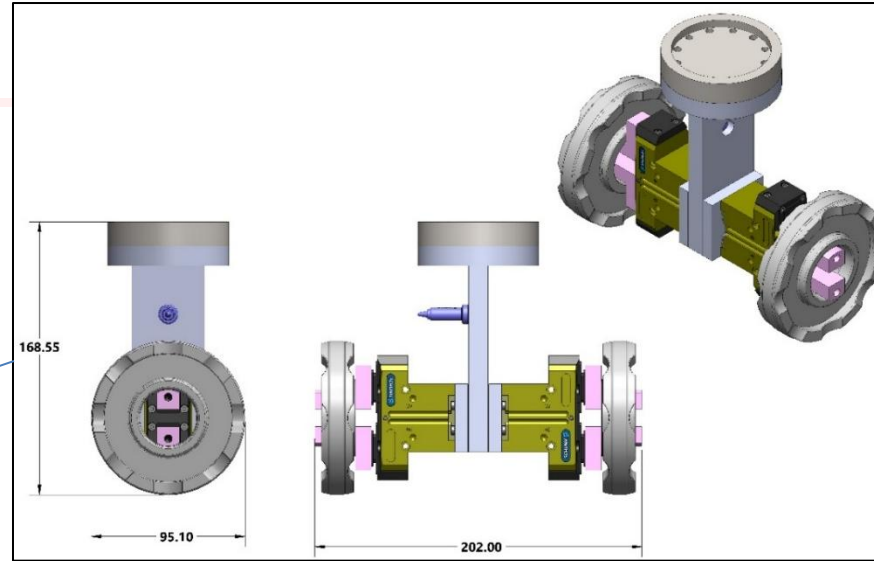
RAD Line-OP30-Iso View



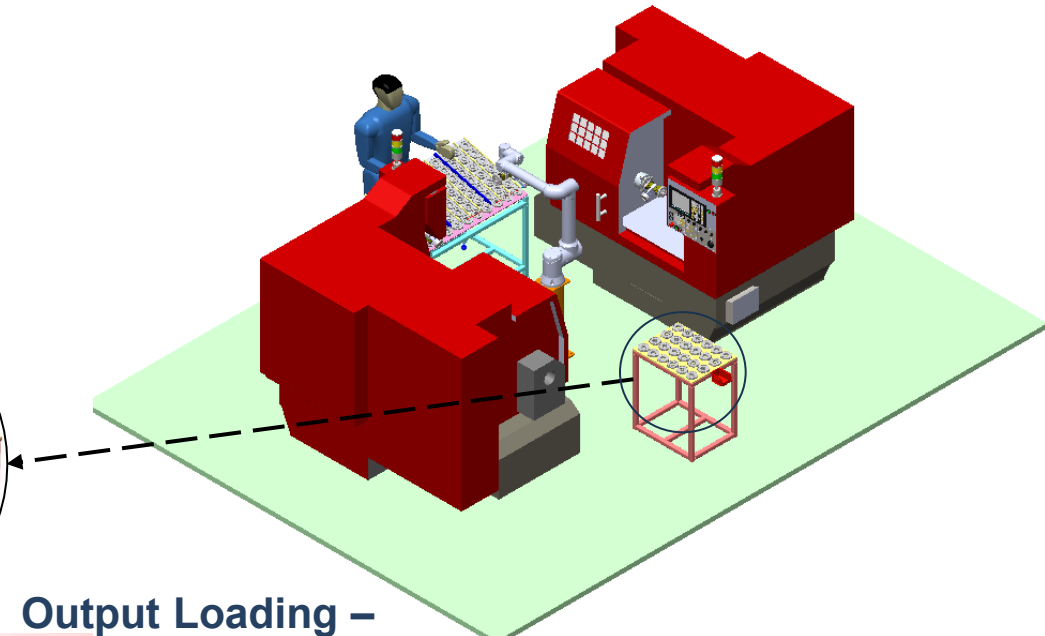
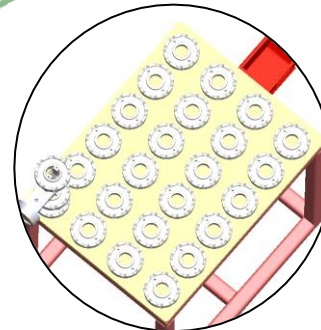
RAD Line–OP30-Gripper Position



RAD Line–OP40-Iso View

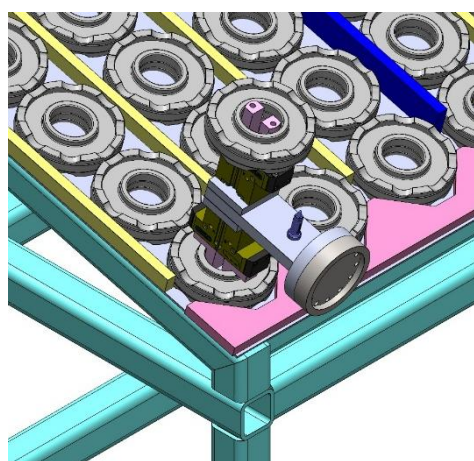


**Input Loading –
Quantity on table : 70Nos**

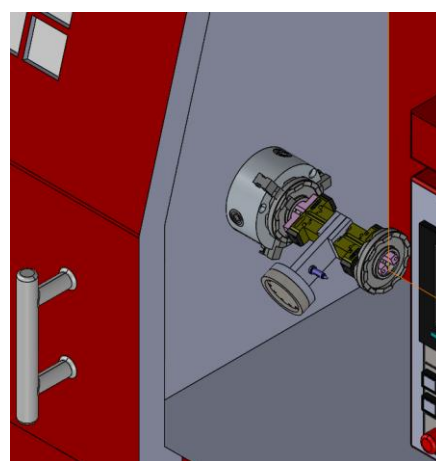
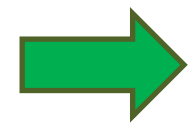


**Output Loading –
Quantity on table : 24Nos**

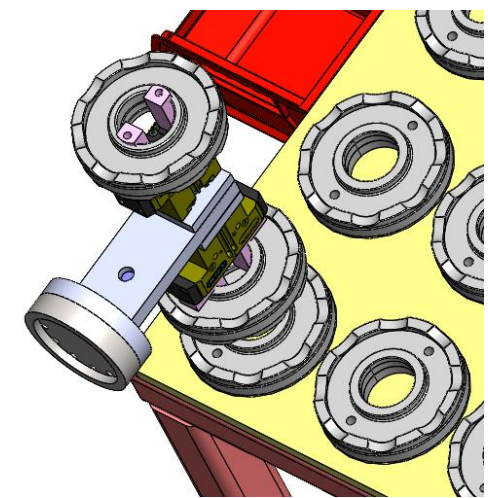
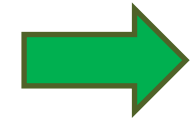
RAD Line–OP40-Gripper Dimension & Position



OP40 - PICKUP



OP40 - LOADING



**Completed OP40 –
LOCATING IN TABLE**

UR12e Product Details



Payload	12.5 kg
Weight (including cable)	33.5 kg
Working Radius	1300mm
Repeatability Accuracy	±0.05mm
Degree of Freedom	6
Programming	12 inch touchscreen with PolyScope graphical user interface
Compliance	EN ISO 13849-1 (PLd Category 3), EN ISO 10218-1

UR12e Product Details

Max Power consumption	615W
Temperature range	0-50°C
IP rating	IP54
Robot installation	Install at any angle
Tool I/O port	Digital In 16 / Digital Out 16 / Analog In 2 / Analog Out 2
Tool I/O power	24V, 2A
Footprint	190mm
Material	Aluminum alloy, PC
Tool Flange	M8 8-pin female, EN ISO-9409-1-50-4-M6

UR12e – Details of Control Cabinet



IP rating	IP44
I/O Port of Electric Control Cabinet	16 digital inputs and outputs, 2 analog inputs or outputs
I/O Power supply for Electric Control Cabinet	24V
Methods of Communication	Modbus TCP (Client/Server), Ethernet/IP Adapter, PROFINET device, PROFIsafe ROS/ROS2
Power Supply	100-240VAC,50-60Hz
Control Box Size	460×449×254 (mm) (W×H×D)
Weight	12kg
Material	Powder Coated Steel

JAKA C 5 - Product Features



Payload	5kg
Weight (including cable)	23kg
Working Radius	954mm
Repeatability Accuracy	±0.05mm
Degree of Freedom	6
Programming	Graphical programming, drag-and-drop programming
Demonstrator Type	Mobile terminal (PAD/mobile phone) APP
Collaborate Operation	Collaborative operations as ISO 10218-1:2011

JAKA C 5 - Physical Properties

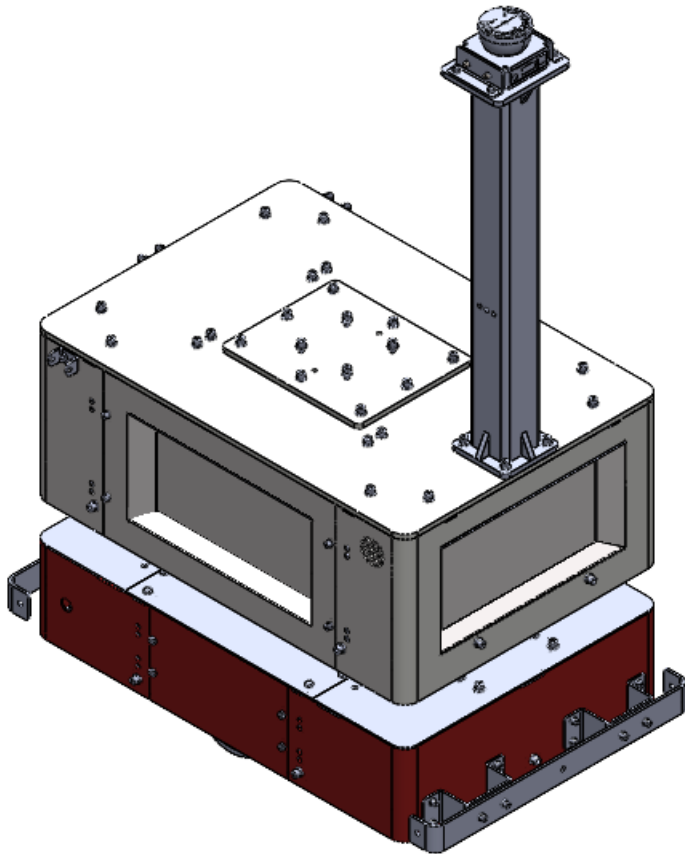
Power consumption	350W
Temperature range	0-50°C
IP rating	IP54
Robot installation	Install at any angle
Tool I/O port	Digital Input-2 / Digital Output-2 / Analog Input-2
Tool I/O power	24V
Base diameter	158mm
Material	Aluminum alloy, PC
Tool I/O size	M8

JAKA C 5 – Detail of control cabinet



IP rating	IP44
I/O Port of Electric Control Cabinet	16 digital inputs and outputs, 2 analog inputs or outputs
I/O Power supply for Electric Control Cabinet	24V
Methods of Communication	TCP/IP, Modbus TCP, Modbus RTU
Power Supply	100-240VAC,50-60Hz
Dimensions of the Electric Control Cabinet	410×307×235 (mm) (W×H×D)
Box Dimensions	600×325×400(mm)(L×W×H)
Weight	12kg
Material	Sprayed carbon steel plate

AIRA – Jack Type AMR



Technical parameter	Detail
• Model	AIRA – JACK SERIES
• Navigation mode	Magnetic Strip / Laser navigation
• Vehicle dimension	1540mm to 2250 (L) *989mm (W) *250mm (H)
• Application	Lifting, Towing, Assembly Operation
• Payload	500 Kg to 3000 Kg
• Lifting Height	60 mm
• Drive mode	Differential speed
• max load	500-3000Kg
• the minimum turning radius	1200-12700mm
• Rotational Diameter	1000mm to 2500mm
• Charging mode	Manual /automatic/ Induction
• Battery	48V/24Ah to 150ah depends on payload Lithium iron phosphate
• Climbing capability	4 (unload , unit : %) /1 (full load , unit : %)
• Communication mode	Wifi (2.4G/5G Dual frequency WIFI Or operator 5G)
• Application	Jack, Lifting, Assembly, Towing, Robot / Cobot Mounting

Gripper In AMR



Make: Zimmer

Technical parameter	Detail
Model	GEP2013IL-00-B
Make	Zimmer Group
Control	IO-Link
Position sensing using process data	Yes
Repetition accuracy +/- [mm]	0.02
Control time [s]	0.055
Voltage [V]	24
Current consumption max. [A]	2
Operating temperature [°C]	5 ... +60
Protection to IEC 60529	IP40
Weight [kg]	0.55

Gripper in Robot



Make: SCHUNK

Technical parameter	Detail
Stroke per jaw [mm]	4
Opening force [N]	300
Opening force (with finger lengths of 0 mm) [N]	340
Min. spring force [N]	65
Weight [kg]	0.24
Recommended workpiece weight [kg]	1.1
Cylinder volume per double stroke [cm ³]	12
Min. operating pressure [bar]	4
Max. operating pressure [bar]	6.5
Nominal operating pressure [bar]	6
Min. air purge pressure [bar]	0.5
Max. air purge pressure [bar]	1
Closing time [s]	0.025
Opening time [s]	0.015
Closing/opening time with spring [s]	0.03
Max. permissible finger length [mm]	75
Max. permissible weight per finger [kg]	0.2
IP protection class	64
Min. ambient temperature [°C]	5
Max. ambient temperature [°C]	90
Repeat accuracy [mm]	0.01
Cleanroom class ISO 14644-1	6
Cleanroom class: ISO 14644-1	2015
Moment Mx max. [Nm]	27
Moment My max. [Nm]	25
Moment Mz max. [Nm]	15
Max. axial force Fz max. [N]	670
Dustproof version	yes

Our Products

**AIRA - Jack
Series**



Payload – 0.5 to 3 tons
**Lifting Height by
60mm**

**AIRA -
Mover Series**



**Payload – 0.5 to 3
tons**
**Lifting Height upto
200mm**

**AIRA – Stacker
Series**



**Payload – 0.5 to 3
tons**
**Lifting Height upto
12 meters**

**AIRA – RT
Series**



Payload – 0.5 to 6 tons
**Lifting Height upto
8 meters**

Material Handling Solutions



Lead Pallet Handling



Pallet Handling



Engine Handling



Tyre Handling



Motor Casing Handling



Assembly Handling

Contact us @

**No.73, Kolathur Village Opposite to Sare Homes,
Palur Road, Singaperumal Kovil, Kanchipuram,
Chennai – 603204.**

info@roboneticssolutions.com

sales@roboneticssolutions.com

+91-6379582983/+91-9791253575 (India)

+971-56-1952902 (Dubai)

<https://www.roboneticssolutions.com/>